

Joseph Telaak

704-351-7396 | jttelaak04@gmail.com | linkedin.com/in/jtelaak

EDUCATION

University of South Carolina

BSE in Computer Engineering, Leadership Distinction in Research

Columbia, SC

Aug. 2022 – Dec. 2024

EXPERIENCE

Chief Executive Officer

Dec. 2024 – Present

Parkeze Inc.

Columbia, SC

- Ported the LoRaWAN gateway stack to Zephyr RTOS with custom high-availability features.
- Developed sensor firmware integrating magnetometer and mmWave radar for vehicle occupancy detection.
- Architected Kafka-based telemetry backend supporting real-time occupancy updates for 1000+ concurrent users.
- Built real-time analytics dashboard with React, GraphQL, and TimescaleDB.
- Created CI/CD pipeline for integration testing and blue/green deployment on Kubernetes.

Research Assistant

Feb. 2023 – Dec. 2024

USC Systems Research on X Lab

Columbia, SC

- Developed firmware for live data streaming from TI IWR series 77–81 GHz mmWave radar SoCs.
- Designed FPGA-based infrastructure for multi-radar synchronization and high-throughput capture.
- Implemented contactless vital sign monitoring via chest-wall vibration analysis validated against ECG.
- Co-authored research on mmWave radar algorithms for 3D vehicle and pedestrian detection.
- Secured research funding through competitive grant proposals.

Project Lead

Jan. 2022 – Feb. 2023

SCGSSM Autonomous Golf Cart

Hartsville, SC

- Engineered ADAS control systems for steering, throttle, and braking actuation.
- Designed custom Nvidia Jetson carrier board with analog video capture and ADC interfacing.
- Implemented real-time computer vision pipeline for obstacle detection and lane tracking.
- Integrated CAN bus and low-level actuator control for drive-by-wire operation.

LEADERSHIP

FIRST Robotics

Columbia, SC

FIRST Technical Advisor, Judge, Robot Inspector

Jan. 2022 – May 2025

SELECTED PROJECTS

LLM Voice Assistant: Far-field mic array with XMOS DSP, Zephyr RTOS on i.MXRT, ESP32 Wi-Fi NIC, multi-chip OTA, and real-time audio streaming to an LLM inference pipeline.

Agentic Audio Streamer: Multi-room audio system using SigmaDSP, DACs, Raspberry Pi CM5, Spotify Connect, and a custom LLM control API.

PiSwitch: 7-port managed router/switch with custom ASIC controller, VLAN support, web UI, and out-of-band management.

Multi-Core Nios II Image Scaler: Multi-core Nios II FPGA image scaler in Quartus with custom DMA fabric, SDRAM frame buffering, and hardware-accelerated bilinear interpolation for real-time video processing.

RISC-V CPU: 5-stage pipelined RISC-V CPU with GPIO peripheral support implemented on FPGA in SystemVerilog.

Pick-n-Place Machine: Desktop PCB assembly machine with OpenPNP vision alignment and automated SMT placement.

SKILLS

Languages: C/C++, Python, Rust, Java, JavaScript/TypeScript, MATLAB, Lua, SystemVerilog

Embedded: Zephyr RTOS, Yocto, Buildroot, FreeRTOS, LoRaWAN, I2C, SPI, UART, CAN, USB

Hardware: Altium, KiCad, Quartus, Platform Designer/Qsys, STM32Cube, MCUXpresso, mmWave Studio, JTAG, Oscilloscope

Systems: Linux, Kubernetes, Docker, Kafka, GraphQL, PostgreSQL, CI/CD, Git

Memberships: IEEE Eta Kappa Nu (HKN), IEEE MTTs, IEEE, ACM, AIAA